

reduced anthropogenic metric values are added to the non-anthropogenic metric values to give the final target metric values.

Example:

Initial Metric Values:

30 35 50 100 80 44 67 88 90 70 50 30 55 90 80 85

Attainment Test: Highest value of metric ≤ 70

Interday Background Level: 40

Interday Rollback Method: Incremental

Out of Attainment Value: 100

Anthropogenic Out of Attainment Value: 60

Anthropogenic Standard: 30 (=70 - 40)

Incremental Reduction Required: 30

Non-Anthropogenic Metric Values:

30 35 40 40 40 40 40 40 40 40 40 30 40 40 40 40

Anthropogenic Metric Values:

0 0 10 60 40 4 27 48 50 30 10 0 15 50 40 45

Reduced Anthropogenic Metric Values:

0 0 0 30 10 0 0 18 20 0 0 0 0 20 10 15

Target Metric Values:

30 35 40 70 50 40 4 58 60 40 40 30 40 60 50 55

A.3.1.3 Interday Rollback - Peak Shaving

To generate target metric values using Peak Shaving rollback, BenMAP simply truncates all anthropogenic metric values at the anthropogenic standard. These reduced anthropogenic metric values are added to the non-anthropogenic metric values to give the final target metric values. **Example:** Initial Metric Values:

30 35 50 100 80 44 67 88 90 70 50 30 55 90 80 85

Attainment Test: Highest value of metric ≤ 70

Interday Background Level: 40

Interday Rollback Method: Peak Shaving

Anthropogenic Standard: 30

Non-Anthropogenic Metric Values:

30 35 40 40 40 40 40 40 40 40 40 30 40 40 40 40

Anthropogenic Metric Values:

0 0 10 60 40 4 27 48 50 30 10 0 15 50 40 45

Reduced Anthropogenic Metric Values:

0 0 10 30 30 4 27 30 30 30 10 0 15 30 30 30

Target Metric Values:

30 35 50 70 70 44 67 70 70 70 50 30 55 70 70 70

A.3.2 Intraday Rollback - Adjusting Hourly Observations

Once a set of target metric values has been calculated for a pollutant with hourly observations (e.g., Ozone), BenMAP must adjust the hourly observations so that they produce the target metric values. There are two supported rollback methods for Intraday Rollback – Percentage and Incremental. Each of these rollback methods requires some preprocessing of the initial monitor observations, and each can require multiple iterations to hit the target metric values.

We will discuss this preprocessing and iteration first, and then go through Percentage and Incremental rollbacks in turn.

For various reasons, each of the Intraday Rollback methods can fail to hit the target metric values during a single pass through the rollback procedure (these will be discussed in detail below). As such, each of the rollback methods uses an iterative approach to get within a threshold of each of the target metric values - currently this threshold is 0.05. The iterative approach works as follows:

For each target metric value, BenMAP calculates the current value of the Attainment Test metric. For the first iteration, the metric value will be calculated using unadjusted hourly observations. For subsequent iterations, the metric value will be calculated using the current values of the adjusted hourly observations.

If the difference between the metric value and the target metric value is less than or equal to 0.05, the rollback procedure is finished. Otherwise, another iteration is required.

The Intraday Background Level specifies the portion of each observation which cannot be affected by human intervention - we call this portion the non-anthropogenic portion. Whatever portion is left over after subtracting out the background level is referred to as the anthropogenic portion. The anthropogenic portion of the initial monitor observations is the only part which will be affected by the Intraday Rollback Method.

In a way analogous to the Interday Rollback procedure, BenMAP calculates the twenty-four hourly anthropogenic observations and the twenty-four hourly non-anthropogenic observations using the following procedure for each hourly observation:

IF the current value of the observation is less than or equal to the Intraday Background Level,

non-anthropogenic observation = observation

anthropogenic observation = 0

ELSE

non-anthropogenic observation = Intraday Background Level

anthropogenic observation = observation - Intraday Background Level

Given (i) an Attainment Test Metric (e.g., EightHourDailyMax), (ii) an Intraday Background Level, and (iii) a target metric value for the day, BenMAP proceeds to adjust hourly observations in the following steps:

1. Calculate the Attainment Test metric (e.g., the 8-hour daily maximum);
2. Identify the “window” - i.e., the set of hours used to calculate the metric (e.g., if the 8-hour daily maximum is achieved in the first 8 hours, then the window is comprised of the first 8 hours);
3. Calculate the non-anthropogenic hourly observations (=min(hourly observation, Intraday Background Level));
4. Calculate the anthropogenic hourly observations (=hourly observation - Intraday Background Level);
5. Calculate the non-anthropogenic metric value (= the metric using the non-anthropogenic hourly observations in the “window”);
6. Calculate the anthropogenic metric value (= the metric using the anthropogenic hourly observations in the “window”);

7. Calculate the anthropogenic target metric value (= the target metric value minus the non- anthropogenic metric value);
8. Calculate the reduction required to get the anthropogenic metric value down to the anthropogenic target metric value;
9. Adjust all anthropogenic hourly observations by the reduction calculated on the previous step;
10. Calculate the adjusted hourly observations (= the adjusted anthropogenic hourly observation + the non-anthropogenic hourly observation).

A.3.2.1 Intraday Rollback - Percentage

Below, we present two examples of a percentage-based Intraday Rollback. In one example, a single iteration is needed, and in the second example, two iterations are required because a number of the monitor values fall below the assumed background level.

A.3.2.1.1 Example: All Hourly Observations Exceed the Intraday Background (Single Iteration)

If all of the hourly observations in a day are greater than the Intraday Background Level, then the above procedure is straightforward and can be accomplished in a single iteration. We illustrate with the following example. Suppose that:

Metric = EightHourDailyMax,

Target metric value for a given day = 85

Intraday Background Level = 40.

And that the hourly observations on that day are:

530	45	50	60	45	45	45	60	70	100	100	100	100
100	100	100	100	60	45	50	45	45	47	47		

Based on these observations, we see that the 8-hour daily maximum = 110.

Assuming a background level of 40, then the Anthropogenic hourly observations are:

490	5	10	20	5	5	5	20	30	60	60	60	60	60
60	60	60	20	5	10	5	5	7	7				

Then, we know:

Anthropogenic metric value = 70.